NAME: -MYDAM ASWITHA PATEL

ROLL NO: - 2520030001

SECTION: -2

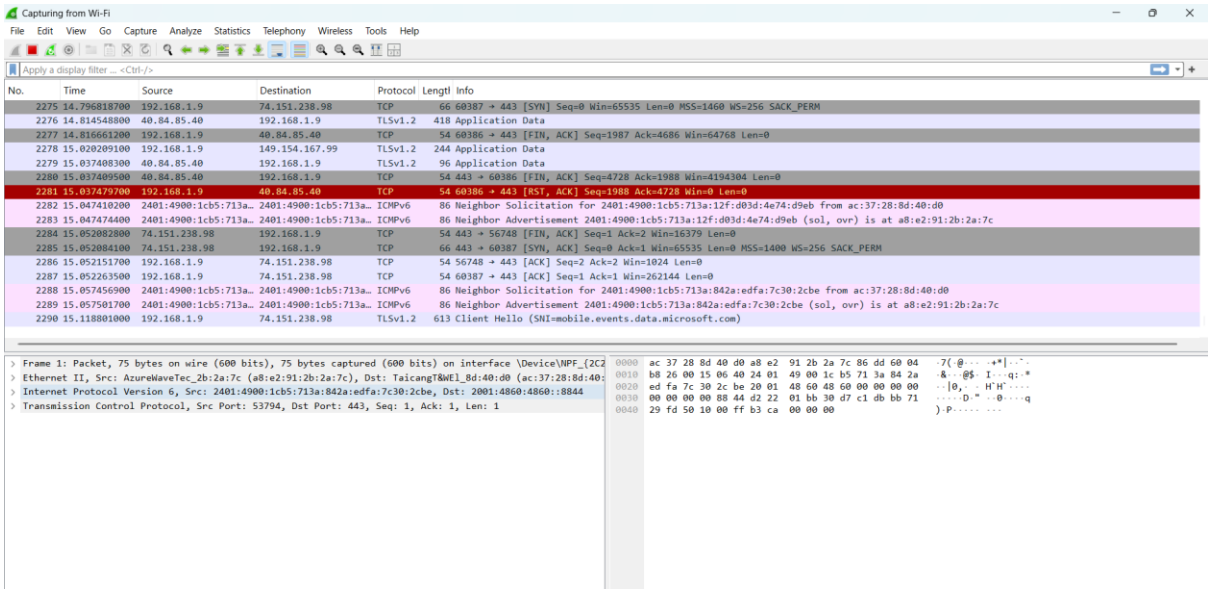
COURSE CODE: -25CS2208E

COMPUTER NETWORKS AND NETWORK PROGARMMING

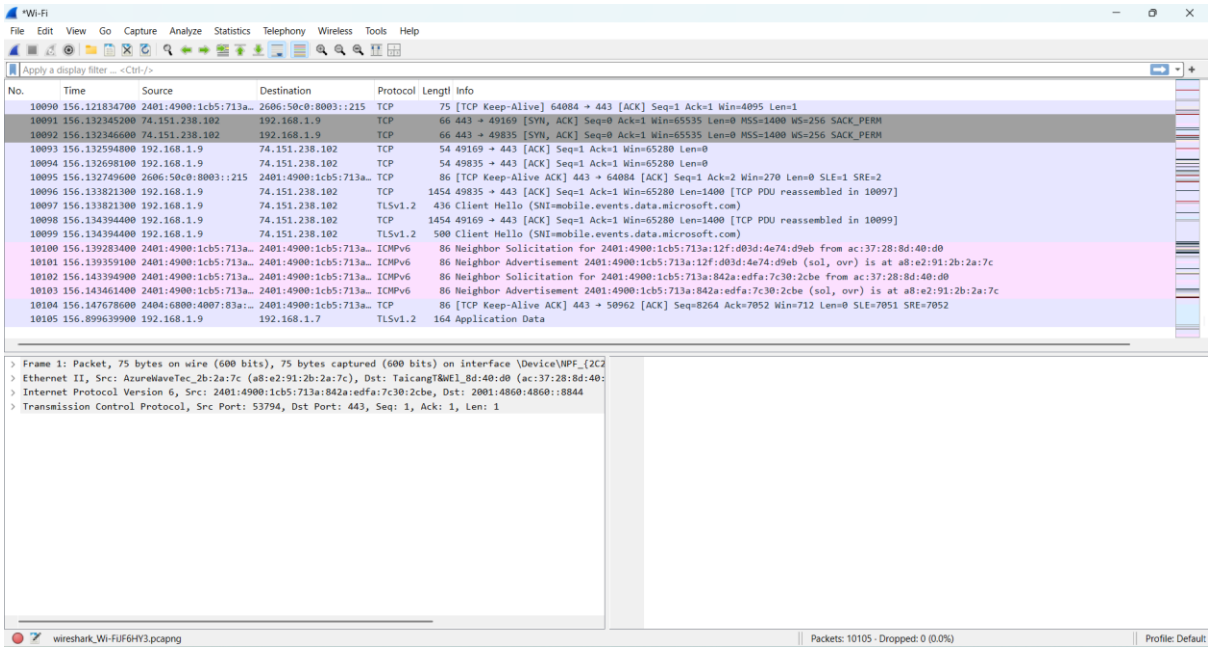
SKILL-1

STEP 1: - Installation of wire shark

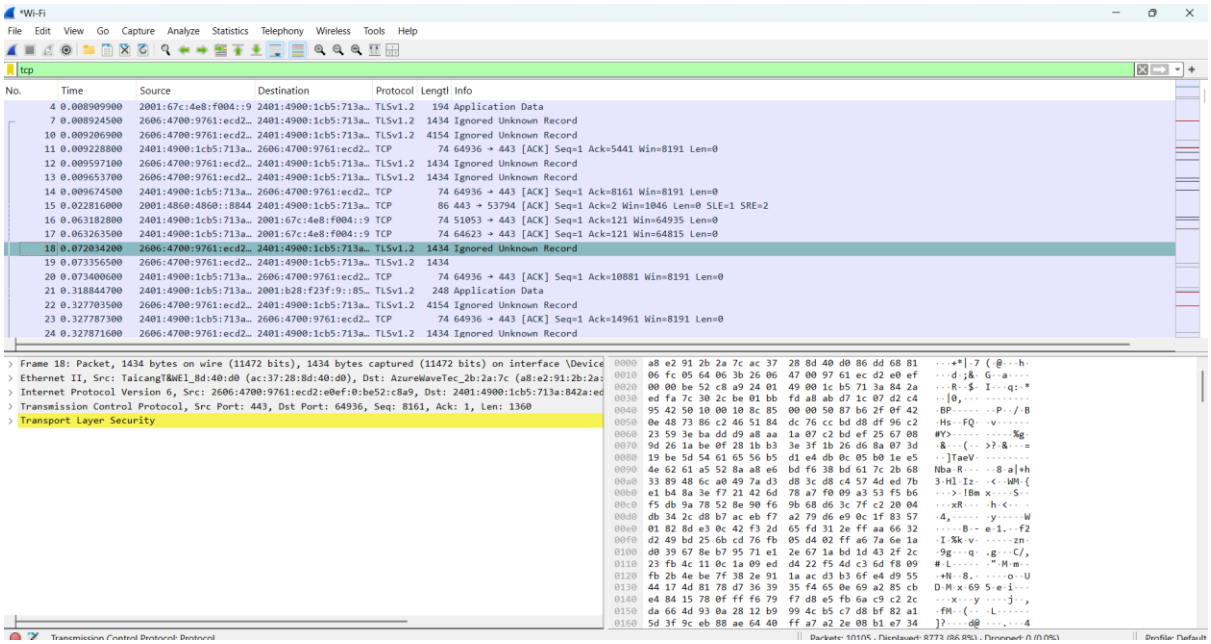
STEP 2: -

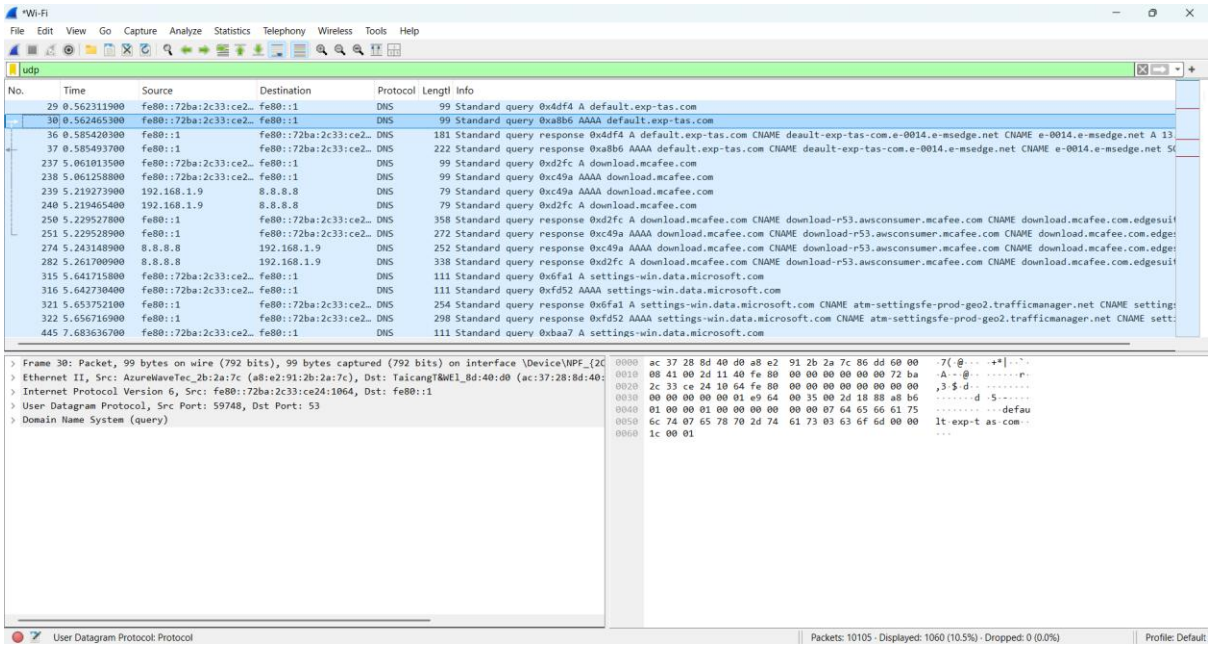


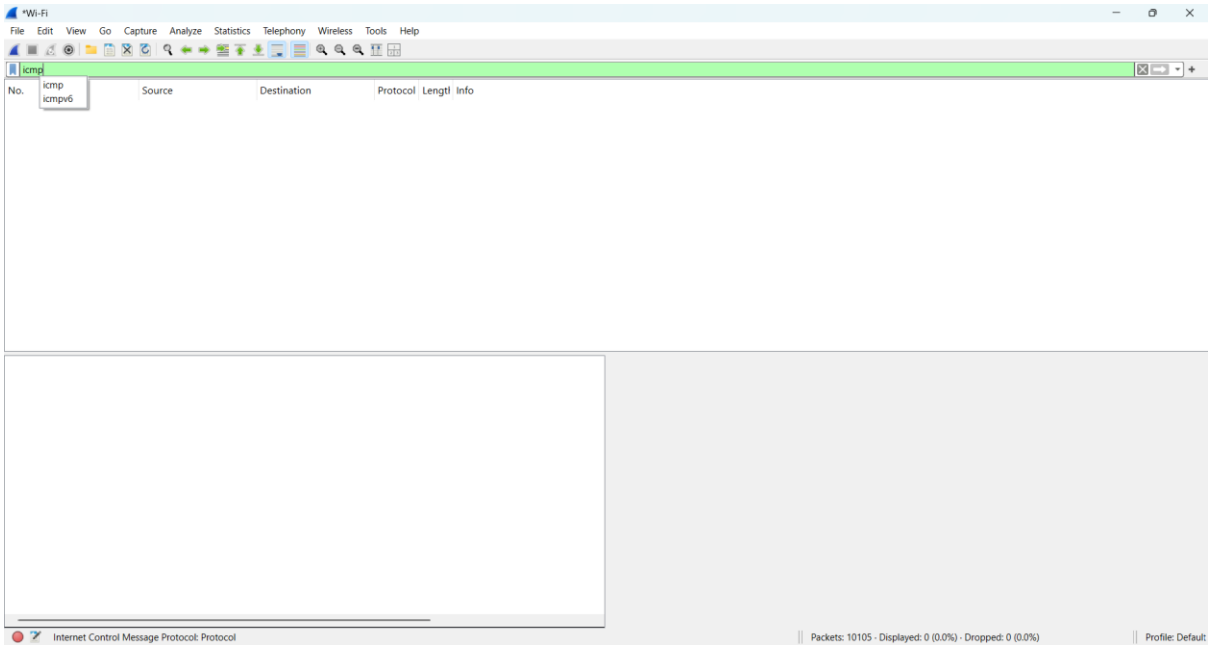
STEP 3: -

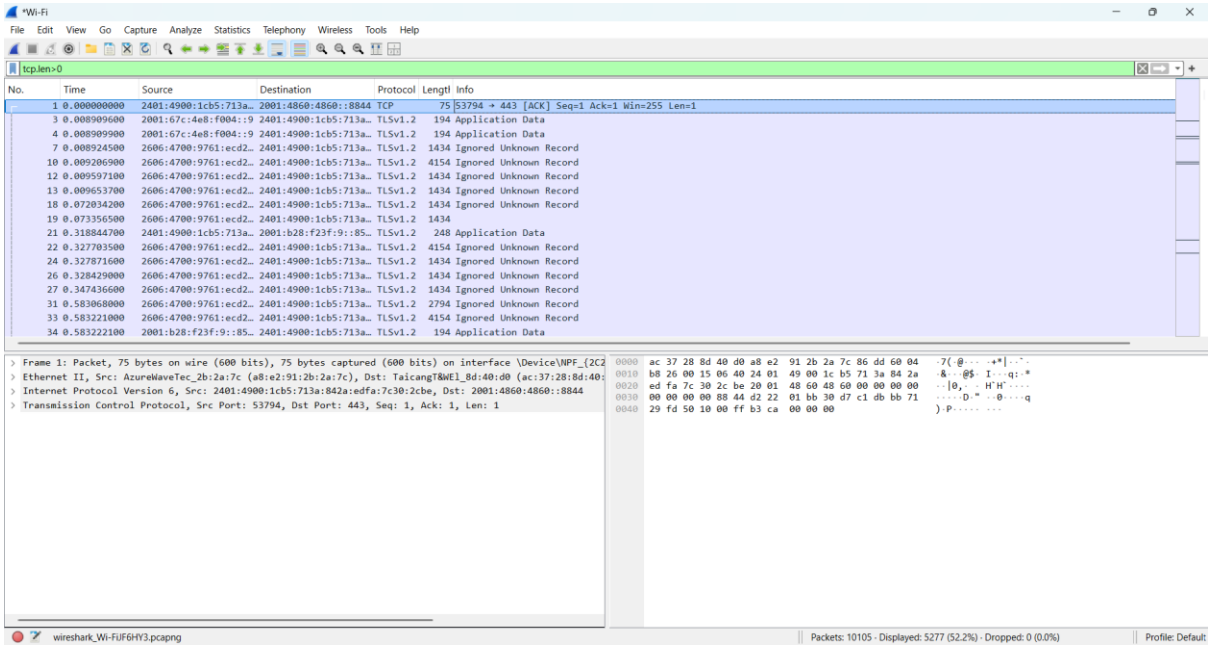


STEP 4: -

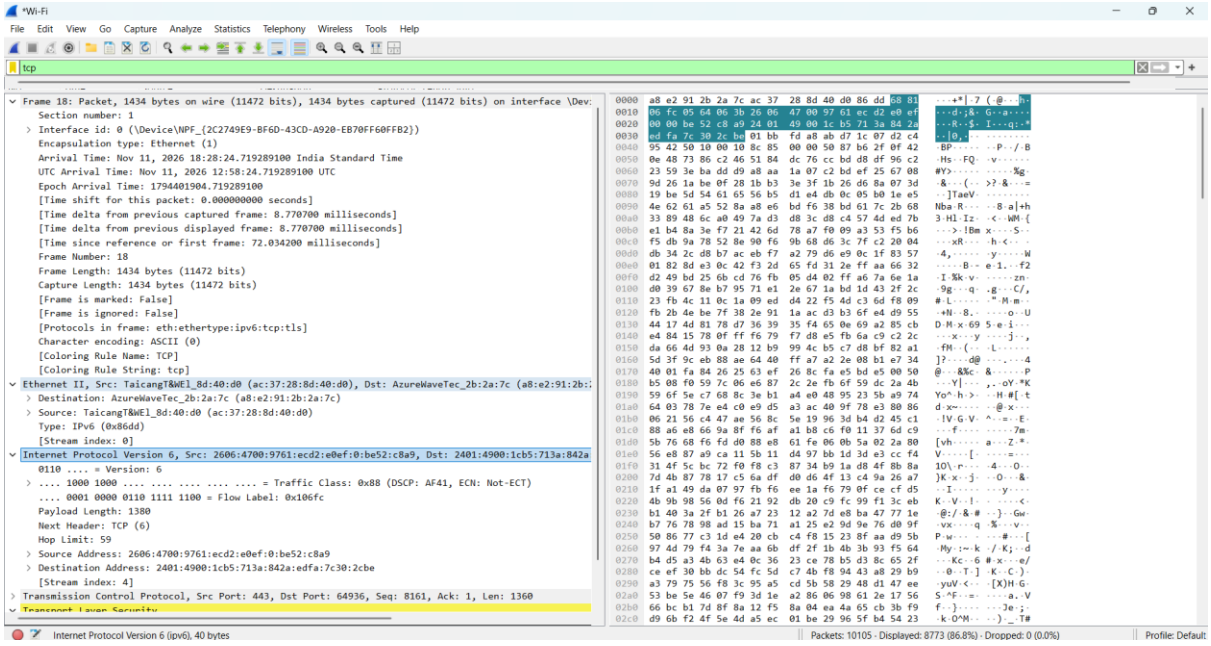

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
4 0.008909900 2001:67c:4e8:f004::9 2401:4900:1cb5:713a… TLSv1.2 194 Application Data
7 0.008924500 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
10 0.009206900 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 4154 Ignored Unknown Record
11 0.009228800 2401:4900:1cb5:713a… 2606:4700:9761:ecd2… TCP 74 64936 → 443 [ACK] Seq=1 Ack=5441 Win=8191 Len=0
12 0.009597100 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
13 0.009653700 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
14 0.009674500 2401:4900:1cb5:713a… 2606:4700:9761:ecd2… TCP 74 64936 → 443 [ACK] Seq=1 Ack=8161 Win=8191 Len=0
15 0.022816000 2001:4860:4860::8844 2401:4900:1cb5:713a… TCP 86 443 → 53794 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
16 0.063182800 2401:4900:1cb5:713a… 2001:67c:4e8:f004::9 TCP 74 51053 → 443 [ACK] Seq=1 Ack=121 Win=64935 Len=0
17 0.063263500 2401:4900:1cb5:713a… 2001:67c:4e8:f004::9 TCP 74 64623 → 443 [ACK] Seq=1 Ack=121 Win=64815 Len=0
18 0.072034200 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
19 0.073356500 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434
20 0.073400600 2401:4900:1cb5:713a… 2606:4700:9761:ecd2… TCP 74 64936 → 443 [ACK] Seq=1 Ack=10881 Win=8191 Len=0
21 0.318844700 2401:4900:1cb5:713a… 2001:b28:f23f:9::85… TLSv1.2 248 Application Data
22 0.327703500 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 4154 Ignored Unknown Record
23 0.327787300 2401:4900:1cb5:713a… 2606:4700:9761:ecd2… TCP 74 64936 → 443 [ACK] Seq=1 Ack=14961 Win=8191 Len=0
24 0.327871600 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
Frame 18: Packet, 1434 bytes on wire (11472 bits), 1434 bytes captured (11472 bits) on interface \Device
Ethernet II, Src: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:d0), Dst: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:
Internet Protocol Version 6, Src: 2606:4700:9761:ecd2:e0ef:0:be52:c8a9, Dst: 2401:4900:1cb5:713a:842a:ed
Transmission Control Protocol, Src Port: 443, Dst Port: 64936, Seq: 8161, Ack: 1, Len: 1360
Transport Layer Security

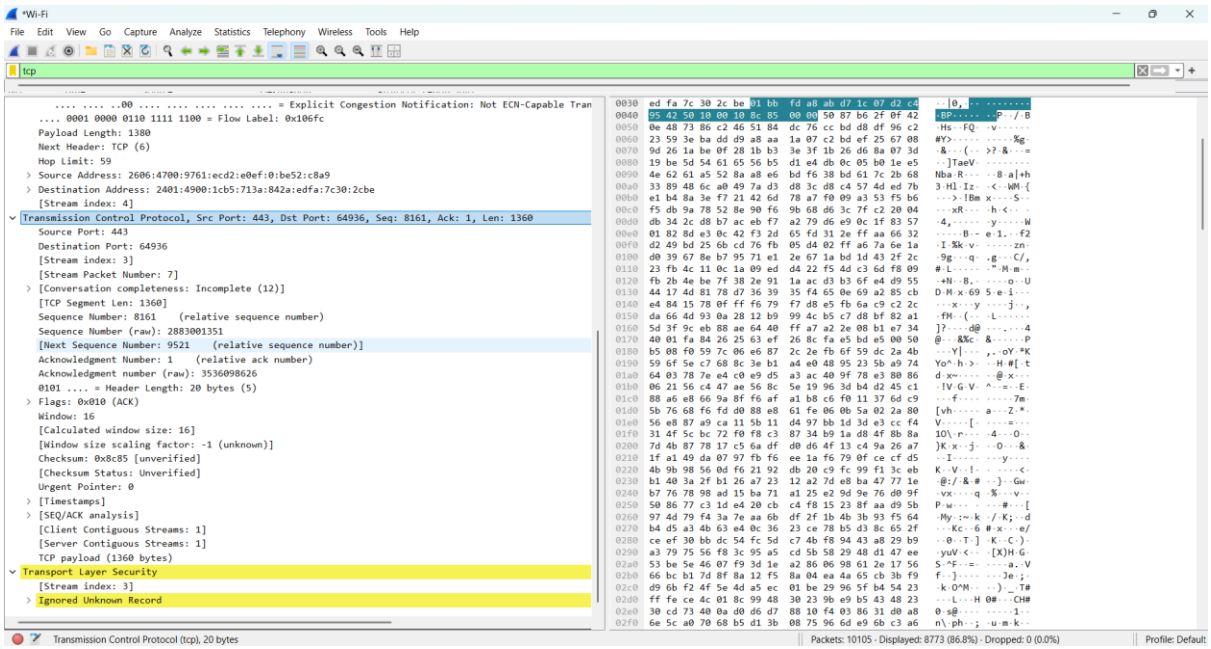

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Lengtl Info
29 0.562311900 fe80::72ba:2c33:ce2… fe80::1 DNS 99 Standard query 0x4df4 A default.exp-tas.com
30 0.562465300 fe80::72ba:2c33:ce2… fe80::1 DNS 99 Standard query 0xa8b6 AAAA default.exp-tas.com
36 0.585420300 fe80::1 fe80::72ba:2c33:ce2… DNS 181 Standard query response 0x4df4 A default.exp-tas.com CNAME deault-exp-tas-com.e-0014.e-msedge.net CNAME e-0014.e-msedge.net A 13
37 0.585493700 fe80::1 fe80::72ba:2c33:ce2… DNS 222 Standard query response 0xa8b6 AAAA default.exp-tas.com CNAME deault-exp-tas-com.e-0014.e-msedge.net CNAME e-0014.e-msedge.net SO
237 5.061013500 fe80::72ba:2c33:ce2… fe80::1 DNS 99 Standard query 0xd2fc A download.mcafee.com
238 5.061258800 fe80::72ba:2c33:ce2… fe80::1 DNS 99 Standard query 0xc49a AAAA download.mcafee.com
239 5.219273900 192.168.1.9 8.8.8.8 DNS 79 Standard query 0xc49a AAAA download.mcafee.com
240 5.219465400 192.168.1.9 8.8.8.8 DNS 79 Standard query 0xd2fc A download.mcafee.com
250 5.229527800 fe80::1 fe80::72ba:2c33:ce2… DNS 358 Standard query response 0xd2fc A download.mcafee.com CNAME download-r53.awsconsumer.mcafee.com CNAME download.mcafee.com.edgesuit
251 5.229528900 fe80::1 fe80::72ba:2c33:ce2… DNS 272 Standard query response 0xc49a AAAA download.mcafee.com CNAME download-r53.awsconsumer.mcafee.com CNAME download.mcafee.com.edges
274 5.243148900 8.8.8.8 192.168.1.9 DNS 252 Standard query response 0xc49a AAAA download.mcafee.com CNAME download-r53.awsconsumer.mcafee.com CNAME download.mcafee.com.edges
282 5.261700900 8.8.8.8 192.168.1.9 DNS 338 Standard query response 0xd2fc A download.mcafee.com CNAME download-r53.awsconsumer.mcafee.com CNAME download.mcafee.com.edgesuit
315 5.641715800 fe80::72ba:2c33:ce2… fe80::1 DNS 111 Standard query 0x6fa1 A settings-win.data.microsoft.com
316 5.642730400 fe80::72ba:2c33:ce2… fe80::1 DNS 111 Standard query 0xfd52 AAAA settings-win.data.microsoft.com
321 5.653752100 fe80::1 fe80::72ba:2c33:ce2… DNS 254 Standard query response 0x6fa1 A settings-win.data.microsoft.com CNAME atm-settingsfe-prod-geo2.trafficmanager.net CNAME setting
322 5.656716900 fe80::1 fe80::72ba:2c33:ce2… DNS 298 Standard query response 0xfd52 AAAA settings-win.data.microsoft.com CNAME atm-settingsfe-prod-geo2.trafficmanager.net CNAME sett
445 7.683636700 fe80::72ba:2c33:ce2… fe80::1 DNS 111 Standard query 0xbaa7 A settings-win.data.microsoft.com
Frame 30: Packet, 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface \Device\NPF_{2C
Ethernet II, Src: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c), Dst: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:
Internet Protocol Version 6, Src: fe80::72ba:2c33:ce24:1064, Dst: fe80::1
User Datagram Protocol, Src Port: 59748, Dst Port: 53
Domain Name System (query)
User Datagram Protocol: Protocol
Packets: 10105 · Displayed: 1060 (10.5%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
icmp
icmpv6
No. Source Destination Protocol Lengtl Info
Internet Control Message Protocol: Protocol
Packets: 10105 · Displayed: 0 (0.0%) · Dropped: 0 (0.0%)
Profile: Default

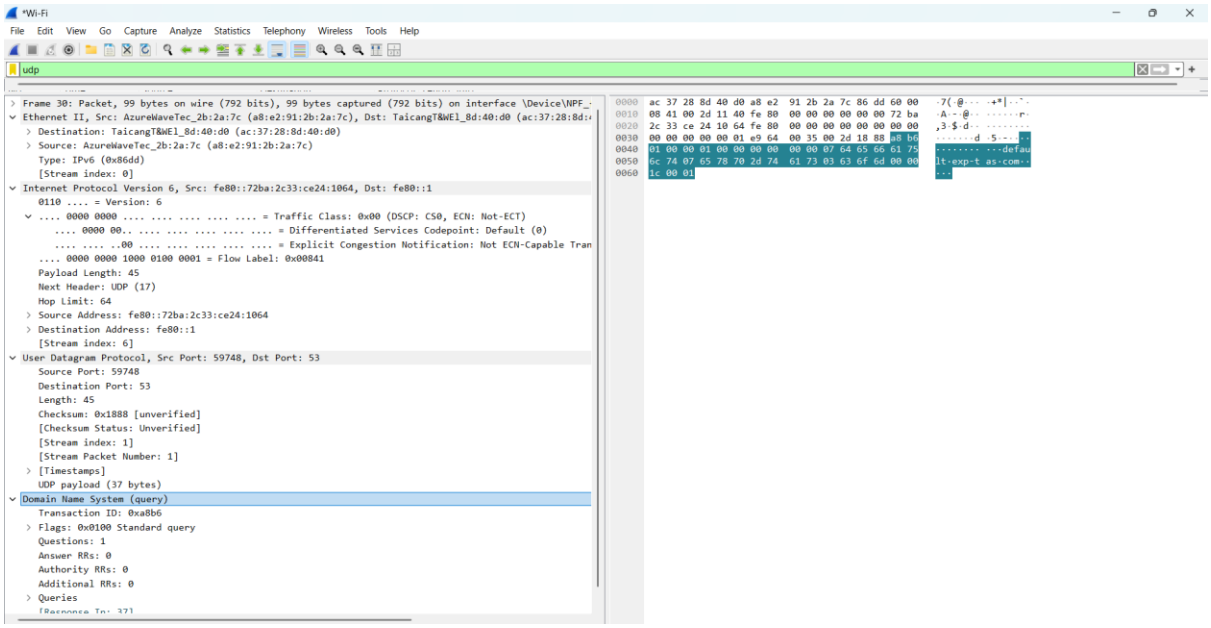

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 2401:4900:1cb5:713a… 2001:4860:4860::8844 TCP 75 53794 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
3 0.008909600 2001:67c:4e8:f004::9 2401:4900:1cb5:713a… TLSv1.2 194 Application Data
4 0.008909900 2001:67c:4e8:f004::9 2401:4900:1cb5:713a… TLSv1.2 194 Application Data
7 0.008924500 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
10 0.009206900 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 4154 Ignored Unknown Record
12 0.009597100 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
13 0.009653700 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
18 0.072034200 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
19 0.073356500 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434
21 0.318844700 2401:4900:1cb5:713a… 2001:b28:f23f:9::85… TLSv1.2 248 Application Data
22 0.327703500 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 4154 Ignored Unknown Record
24 0.327871600 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
26 0.328429000 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
27 0.347436600 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 1434 Ignored Unknown Record
31 0.583068000 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 2794 Ignored Unknown Record
33 0.583221000 2606:4700:9761:ecd2… 2401:4900:1cb5:713a… TLSv1.2 4154 Ignored Unknown Record
34 0.583222100 2001:b28:f23f:9::85… 2401:4900:1cb5:713a… TLSv1.2 194 Application Data
Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{2C2
Ethernet II, Src: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c), Dst: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:
Internet Protocol Version 6, Src: 2401:4900:1cb5:713a:842a:edfa:7c30:2cbe, Dst: 2001:4860:4860::8844
Transmission Control Protocol, Src Port: 53794, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
0000 ac 37 28 8d 40 d0 a8 e2 91 2b 2a 7c 86 dd 60 04
0010 b8 26 00 15 06 40 24 01 49 00 1c b5 71 3a 84 2a
0020 ed fa 7c 30 2c be 20 01 48 60 48 60 00 00 00 00
0030 00 00 00 00 88 44 d2 22 01 bb 30 d7 c1 db bb 71
0040 29 fd 50 10 00 ff b3 ca 00 00 00
wireshark_Wi-FiJF6HY3.pcapng
Packets: 10105 · Displayed: 5277 (52.2%) · Dropped: 0 (0.0%)
Profile: Default

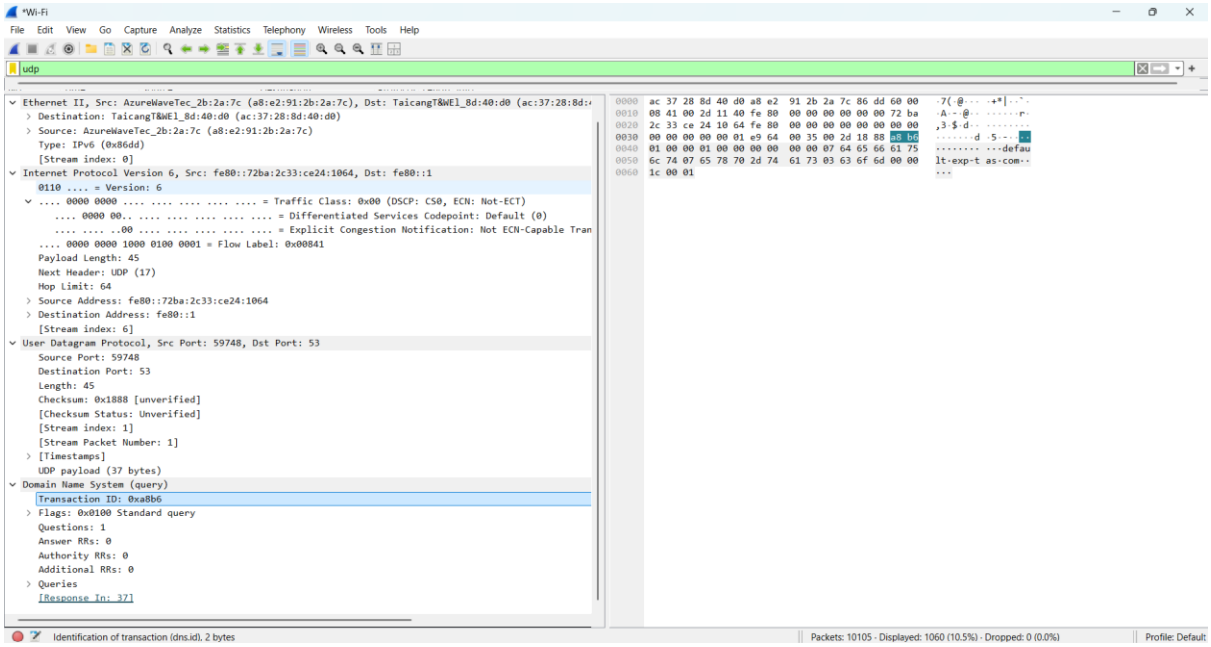
STEP 5: - TCP


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
Frame 18: Packet, 1434 bytes on wire (11472 bits), 1434 bytes captured (11472 bits) on interface \Dev
Section number: 1
Interface id: 0 (\Device\NPF_{2C2749E9-BF6D-43CD-A920-EB70FF60FFB2})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 18:28:24.719289100 India Standard Time
UTC Arrival Time: Nov 11, 2026 12:58:24.719289100 UTC
Epoch Arrival Time: 1794401904.719289100
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 8.770700 milliseconds]
[Time delta from previous displayed frame: 8.770700 milliseconds]
[Time since reference or first frame: 72.034200 milliseconds]
Frame Number: 18
Frame Length: 1434 bytes (11472 bits)
Capture Length: 1434 bytes (11472 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:d0), Dst: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:
Destination: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c)
Source: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:d0)
Type: IPv6 (0x86dd)
[Stream index: 0]
Internet Protocol Version 6, Src: 2606:4700:9761:ecd2:e0ef:0:be52:c8a9, Dst: 2401:4900:1cb5:713a:842a
0110 .... = Version: 6
.... 1000 1000 .... = Traffic Class: 0x88 (DSCP: AF41, ECN: Not-ECT)
.... 0001 0000 0110 1111 1100 = Flow Label: 0x106fc
Payload Length: 1380
Next Header: TCP (6)
Hop Limit: 59
Source Address: 2606:4700:9761:ecd2:e0ef:0:be52:c8a9
Destination Address: 2401:4900:1cb5:713a:842a:edfa:7c30:2cbe
[Stream index: 4]
Transmission Control Protocol, Src Port: 443, Dst Port: 64936, Seq: 8161, Ack: 1, Len: 1360
Transport Layer Security
Internet Protocol Version 6 (ipv6), 40 bytes
Packets: 10105 · Displayed: 8773 (86.8%) · Dropped: 0 (0.0%)
Profile: Default

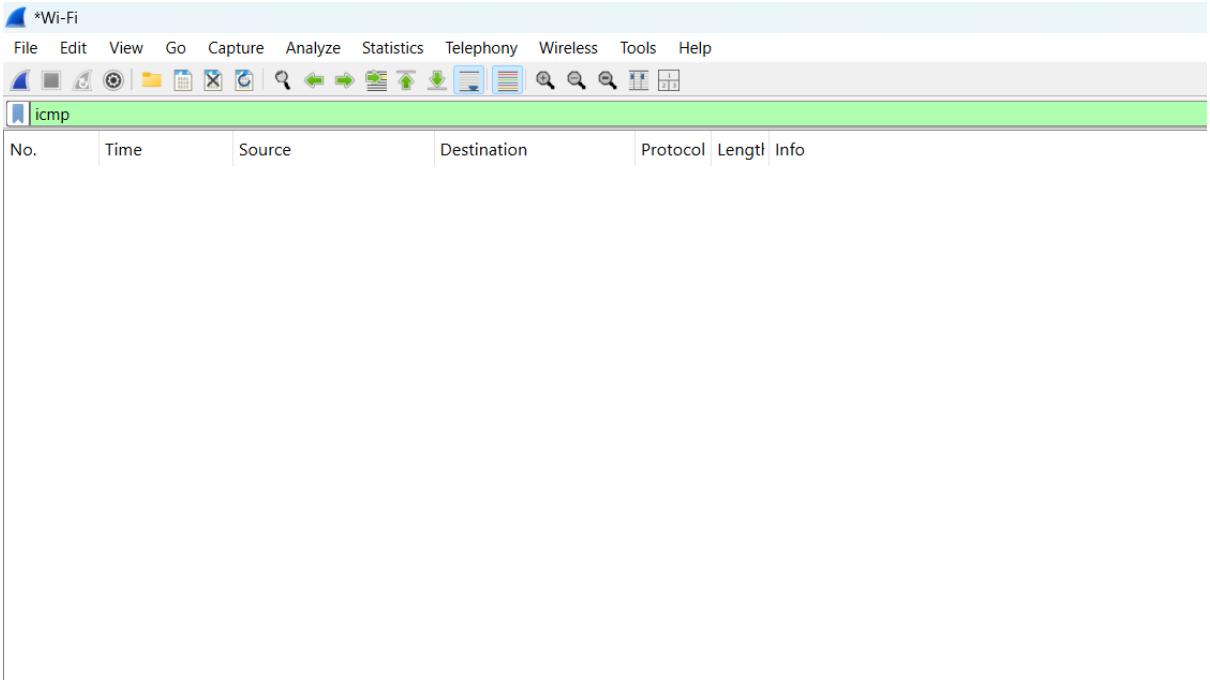

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
....00 .... = Explicit Congestion Notification: Not ECN-Capable Tran
.... 0001 0000 0110 1111 1100 = Flow Label: 0x106fc
Payload Length: 1380
Next Header: TCP (6)
Hop Limit: 59
Source Address: 2606:4700:9761:ecd2:e0ef:0:be52:c8a9
Destination Address: 2401:4900:1cb5:713a:842a:edfa:7c30:2cbe
[Stream index: 4]
Transmission Control Protocol, Src Port: 443, Dst Port: 64936, Seq: 8161, Ack: 1, Len: 1360
Source Port: 443
Destination Port: 64936
[Stream index: 3]
[Stream Packet Number: 7]
[Conversation completeness: Incomplete (12)]
[TCP Segment Len: 1360]
Sequence Number: 8161 (relative sequence number)
Sequence Number (raw): 2883001351
[Next Sequence Number: 9521 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3536098626
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 16
[Calculated window size: 16]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x8c85 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1360 bytes)
Transport Layer Security
[Stream index: 3]
Ignored Unknown Record
Transmission Control Protocol (tcp), 20 bytes
Packets: 10105 · Displayed: 8773 (86.8%) · Dropped: 0 (0.0%)
Profile: Default

UDP

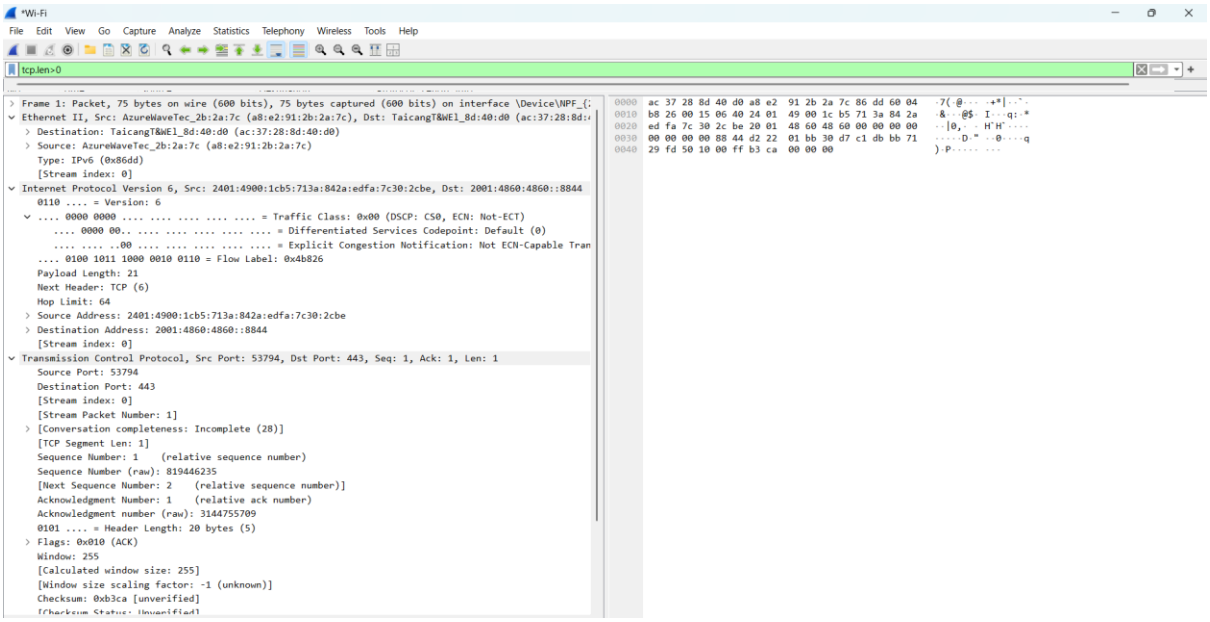

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Frame 30: Packet, 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface \Device\NPF_
Ethernet II, Src: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c), Dst: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:
Destination: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:d0)
Source: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c)
Type: IPv6 (0x86dd)
[Stream index: 0]
Internet Protocol Version 6, Src: fe80::72ba:2c33:ce24:1064, Dst: fe80::1
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
.... 0000 00.. .... = Differentiated Services Codepoint: Default (0)
....00 .... = Explicit Congestion Notification: Not ECN-Capable Tran
.... 0000 0000 1000 0100 0001 = Flow Label: 0x00841
Payload Length: 45
Next Header: UDP (17)
Hop Limit: 64
Source Address: fe80::72ba:2c33:ce24:1064
Destination Address: fe80::1
[Stream index: 6]
User Datagram Protocol, Src Port: 59748, Dst Port: 53
Source Port: 59748
Destination Port: 53
Length: 45
Checksum: 0x1888 [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (37 bytes)
Domain Name System (query)
Transaction ID: 0xa8b6
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
0000 ac 37 28 8d 40 d0 a8 e2 91 2b 2a 7c 86 dd 60 00
0010 08 41 00 2d 11 40 fe 80 00 00 00 00 00 00 72 ba
0020 2c 33 ce 24 10 64 fe 80 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 01 e9 64 00 35 00 2d 18 88 a8 b6
0040 01 00 00 01 00 00 00 00 00 00 07 64 65 66 61 75
0050 6c 74 07 65 78 70 2d 74 61 73 03 63 6f 6d 00 00
0060 1c 00 01


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Ethernet II, Src: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c), Dst: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:
Destination: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:d0)
Source: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c)
Type: IPv6 (0x86dd)
[Stream index: 0]
Internet Protocol Version 6, Src: fe80::72ba:2c33:ce24:1064, Dst: fe80::1
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
.... 0000 00.. .... = Differentiated Services Codepoint: Default (0)
....00 .... = Explicit Congestion Notification: Not ECN-Capable Tran
.... 0000 0000 1000 0100 0001 = Flow Label: 0x00841
Payload Length: 45
Next Header: UDP (17)
Hop Limit: 64
Source Address: fe80::72ba:2c33:ce24:1064
Destination Address: fe80::1
[Stream index: 6]
User Datagram Protocol, Src Port: 59748, Dst Port: 53
Source Port: 59748
Destination Port: 53
Length: 45
Checksum: 0x1888 [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (37 bytes)
Domain Name System (query)
Transaction ID: 0xa8b6
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
[Response In: 37]
0000 ac 37 28 8d 40 d0 a8 e2 91 2b 2a 7c 86 dd 60 00
0010 08 41 00 2d 11 40 fe 80 00 00 00 00 00 00 72 ba
0020 2c 33 ce 24 10 64 fe 80 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 01 e9 64 00 35 00 2d 18 88 a8 b6
0040 01 00 00 01 00 00 00 00 00 00 07 64 65 66 61 75
0050 6c 74 07 65 78 70 2d 74 61 73 03 63 6f 6d 00 00
0060 1c 00 01
Identification of transaction (dns.id), 2 bytes
Packets: 10105 · Displayed: 1060 (10.5%) · Dropped: 0 (0.0%)
Profile: Default

ICMP


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Lengtl Info

TCP.LEN>0


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{
Ethernet II, Src: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c), Dst: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:
Destination: TaicangT&WEl_8d:40:d0 (ac:37:28:8d:40:d0)
Source: AzureWaveTec_2b:2a:7c (a8:e2:91:2b:2a:7c)
Type: IPv6 (0x86dd)
[Stream index: 0]
Internet Protocol Version 6, Src: 2401:4900:1cb5:713a:842a:edfa:7c30:2cbe, Dst: 2001:4860:4860::8844
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
.... 0000 00.. .... = Differentiated Services Codepoint: Default (0)
....00 .... = Explicit Congestion Notification: Not ECN-Capable Tran
.... 0100 1011 1000 0010 0110 = Flow Label: 0x4b826
Payload Length: 21
Next Header: TCP (6)
Hop Limit: 64
Source Address: 2401:4900:1cb5:713a:842a:edfa:7c30:2cbe
Destination Address: 2001:4860:4860::8844
[Stream index: 0]
Transmission Control Protocol, Src Port: 53794, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
Source Port: 53794
Destination Port: 443
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (28)]
[TCP Segment Len: 1]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 819446235
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3144755709
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 255]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xb3ca [unverified]
0000 ac 37 28 8d 40 d0 a8 e2 91 2b 2a 7c 86 dd 60 04
0010 b8 26 00 15 06 40 24 01 49 00 1c b5 71 3a 84 2a
0020 ed fa 7c 30 2c be 20 01 48 60 48 60 00 00 00 00
0030 00 00 00 00 88 44 d2 22 01 bb 30 d7 c1 db bb 71
0040 29 fd 50 10 00 ff b3 ca 00 00 00

```
Sequence Number: 1    (relative sequence number)
Sequence Number (raw): 819446235
[Next Sequence Number: 2    (relative sequence number)]
Acknowledgment Number: 1    (relative ack number)
Acknowledgment number (raw): 3144755709
0101 .... = Header Length: 20 bytes (5)
> Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 255]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xb3ca [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> [Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1 byte)
TCP segment data (1 byte)
```

STEP 6: - TCP PACKET

Given:

Ethernet Header = 14 bytes

IPv6 Header = 40 bytes

TCP Header = 20 bytes

Application Data (TCP Payload) = 1360 bytes

Frame Length = 1434 bytes

A. Total Header Overhead:

Total Header Overhead = Ethernet Header + IPv6 Header + TCP Header

= 14 + 40 + 20

= **74 bytes**

B. Total Packet Size:

Total Packet Size = Header Overhead + Application Data

= 74 + 1360

= **1434 bytes**

C. Payload Efficiency:

Payload Efficiency = (Application Data / Total Frame Length) × 100

= (1360 / 1434) × 100

= **94.84%**

D. Header Overhead Percentage:

Header Overhead Percentage = (Total Header Overhead / Total Frame Length) × 100

= (74 / 1434) × 100

= **5.16%**

STEP 7: UDP PACKET

Given:

Ethernet Header = 14 bytes

IPv6 Header = 40 bytes

UDP Header = 8 bytes

Application Data (UDP Payload) = 37 bytes

Frame Length = 99 bytes

A. Total Header Overhead:

Total Header Overhead = Ethernet Header + IPv6 Header + UDP Header

= 14 + 40 + 8

= **62 bytes**

B. Total Packet Size:

Total Packet Size = Header Overhead + Application Data

= 62 + 37

= **99 bytes**

C. Payload Efficiency:

Payload Efficiency = (Application Data / Total Frame Length) × 100

= (37 / 99) × 100

= **37.37%**

D. Header Overhead Percentage:

Header Overhead Percentage = (Total Header Overhead / Total Frame Length) × 100

= (62 / 99) × 100

= **62.63%**

STEP 8: -

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	1434 bytes	99 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	40 bytes (IPv6)	40 bytes (IPv6)
Transport Header Length	20 bytes (TCP)	8 bytes (UDP)
IP Total Length	1420 bytes	85 bytes
Header Overhead	74 bytes	62 bytes
Payload Length	1360 bytes	37 bytes
Payload Efficiency	94.84%	37.37%
Header Overhead %	5.16%	62.63%

RESULT:

The TCP packet has a payload efficiency of 94.84%, whereas the UDP packet has a payload efficiency of 37.37%.

TCP carries a larger application payload in the selected frame, while UDP has a smaller payload in this particular capture.